

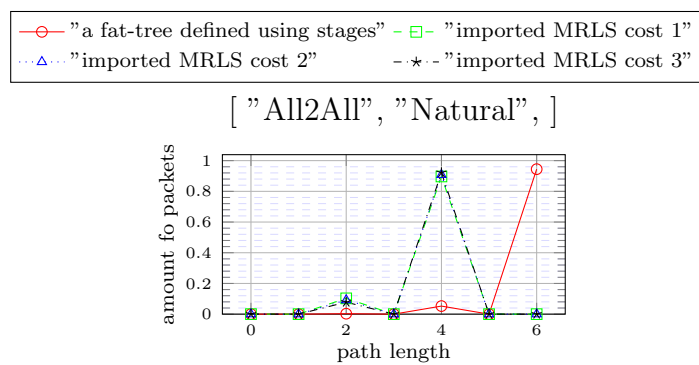


Figure 1: X: ["All2All", "Natural",]

The following versions used in the simulations.

- heads/alex-dev-14138fe7cccb5c3a78a50ccfa156c15e0db90f1e(0.6.3)
- heads/alex-dev-ef489f87de796fb3ec38c1bb4368719b2cb116eb(0.6.3)
- heads/alex-dev-7775ae14e7573de7f634e7a6d77d154c320a8b9a(0.6.3)
- heads/alex-stable-0ddd70b98d380a94f1ddc39d4b3992e370e55d3d(0.6.3)

```
Configuration{
  random_seed: ![ 42, 935, 128643 ],
  warmup: 999999999999,
  measured: 999999999999,
  general_frequency_divisor: 2,
  topology: topo![
    MultiStage{
      stages: [
        Fat{ bottom_factor: 18, top_factor: 18 },
        Fat{ bottom_factor: 18, top_factor: 18 },
        Fat{ bottom_factor: 18, top_factor: 18 }],
      servers_per_leaf: 18,
      legend_name: "a fat-tree defined using stages"},
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_5832_18_2916_36" }],
      servers_per_leaf: 18,
      legend_name: "imported MRLS cost 1"},
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_8748_24_5832_36" }],
      servers_per_leaf: 12,
      legend_name: "imported MRLS cost 2"},
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_11664_27_8748_36" }],
      servers_per_leaf: 9,
      legend_name: "imported MRLS cost 3"}],
  traffic: ![
    TrafficMap{
      tasks: 104976,
      map: ![
        Identity{ legend_name: "Natural" }],
      application: ![
        Burst{
          tasks: 104976,
          messages_per_task: 104975,
          expected_messages_to_consume_per_task: 104975,
          pattern: ElementComposition{
            pattern: CartesianTransform{
              sides: [ 104976 ],
              shift: [ 1 ]}},
          message_size: 1}],
          legend_name: "All2All"}],
        maximum_packet_size: 16,
        router: InputOutput{
          virtual_channels: topo![ 3, 4, 4, 4 ],
          virtual_channel_policies: topo![
            [
              OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
              OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
              LowestLabel,
              EnforceFlowControl,
              Random],
            [
              MapHop{
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                  ArgumentVC{
                    allowed: [ 0 ]},
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                  ArgumentVC{
                    allowed: [ 1 ]},
                  ArgumentVC{
                    allowed: [ 2 ]},
                  ArgumentVC{
                    allowed: [ 2 ]},
                  ArgumentVC{
                    allowed: [ 3 ]},
                  ArgumentVC{
                    allowed: [ 3 ]}],
                  VecLabel{
                    label_vector: [ 0, 64, 80 ]},
                  OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
                  OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
                  LowestLabel,
                  EnforceFlowControl,
                  Random],
                [
                  MapHop{
                    hop_to_policy: [
                      ArgumentVC{
                        allowed: [ 0 ]},
                      ArgumentVC{
                        allowed: [ 0 ]},
                      ArgumentVC{
                        allowed: [ 1 ]},
                    ]}
                  ]
                ]
              ]
            ]
          ]
        ]
      ]
    ]
  ]
}
```

```

    ArgumentVC{
      allowed: [ 1 ]},
    ArgumentVC{
      allowed: [ 2 ]},
    ArgumentVC{
      allowed: [ 2 ]},
    ArgumentVC{
      allowed: [ 3 ]},
    ArgumentVC{
      allowed: [ 3 ]}],
  VecLabel{
    label_vector: [ 0, 64, 80 ]},
  OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
  OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
  LowestLabel,
  EnforceFlowControl,
  Random],
[
  MapHop{
    hop_to_policy: [
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      ArgumentVC{
        allowed: [ 0 ]},
      ArgumentVC{
        allowed: [ 1 ]},
      ArgumentVC{
        allowed: [ 1 ]},
      ArgumentVC{
        allowed: [ 2 ]},
      ArgumentVC{
        allowed: [ 2 ]},
      ArgumentVC{
        allowed: [ 3 ]},
      ArgumentVC{
        allowed: [ 3 ]}],
    VecLabel{
      label_vector: [ 0, 64, 80 ]},
    OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
    OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
    LowestLabel,
    EnforceFlowControl,
    Random]],
  allocator: Random,
  buffer_size: 128,
  bubble: false,
  flit_size: 16,
  intransit_priority: false,
  allow_request_busy_port: true,
  output_buffer_size: 64,
  crossbar_frequency_divisor: 1,
  crossbar_delay: 2},
routing: topo![
  UpDown{ legend_name: "minimal routing" },
  Polarized{ include_labels: true, legend_name: "Polarized original" },
  Polarized{ include_labels: true, legend_name: "Polarized original" },
  Polarized{ include_labels: true, legend_name: "Polarized original" }],
link_classes: [
  LinkClass{ delay: 2 },
  LinkClass{ delay: 2 },
  LinkClass{ delay: 2 },
  LinkClass{ delay: 2 }],
launch_configurations: [
  Slurm{ job_pack_size: 1, time: "50-00:00:00" }]]

```